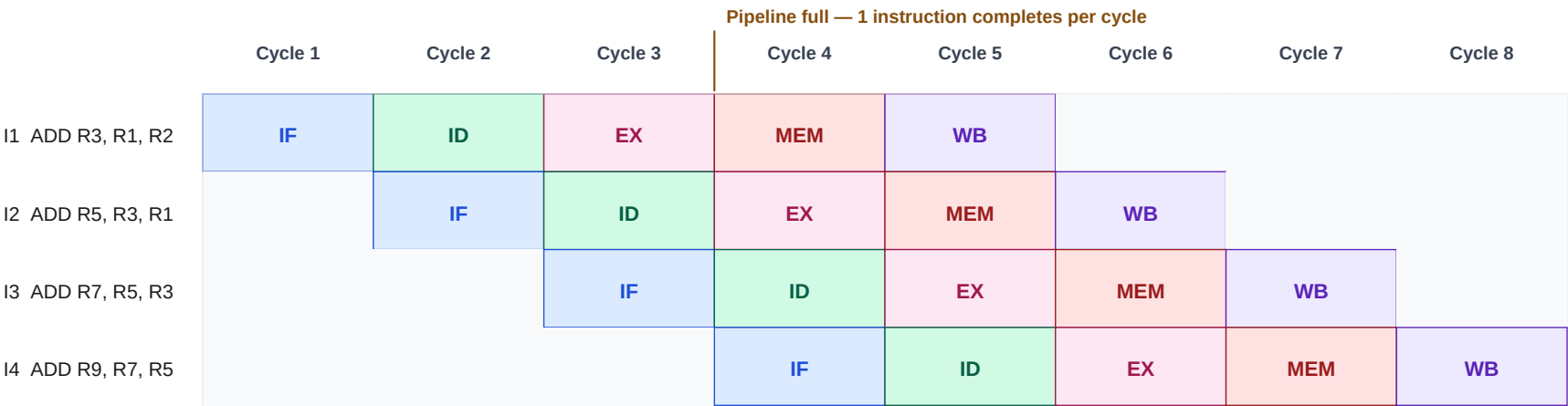


Pipeline timing diagram

Four instructions progressing through 5 pipeline stages. One instruction completes every cycle once the pipeline is full.



Stage key:



Data hazards (e.g. I2 reading R3 written by I1) are resolved by the forwarding unit — zero stall cycles. A load-use hazard inserts 1 bubble stall. Branch mispredictions flush 2 stages.

Pipeline datapath

Components in each stage connected by pipeline registers. Data flows left to right; writeback loops back to register file.

